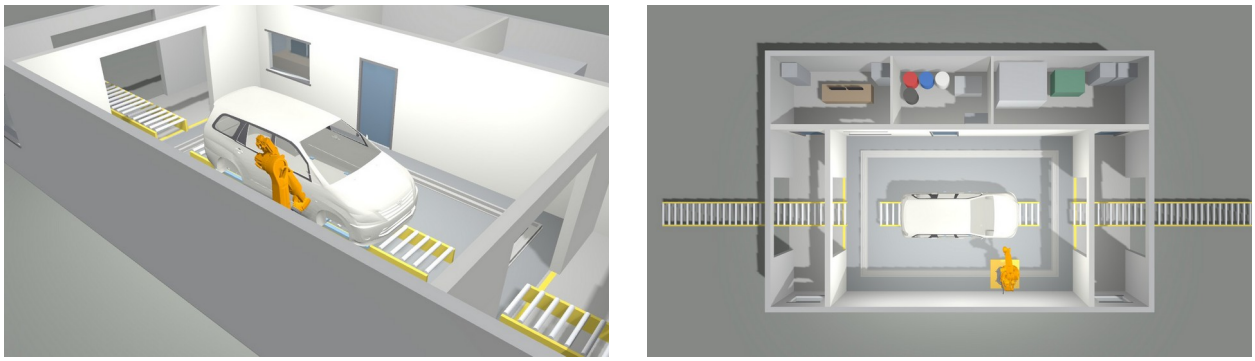
Robotic Car Painting Cell - Simulation

Technical report

Software: ROS 2 Jazzy, MoveIt 2, Gazebo Harmonic, SolidWorks Robot: KUKA KR 210 R3100-2 on a 2-axis track

1. Introduction

The goal of this project is a simulated painting cell in which one industrial robot paints the exterior of a car body (about 4.8 x 1.9 x 1.6 m). The car body enters the booth on a skid conveyor, the robot travels around it on a floor track and paints six surfaces - front, hood, roof, both sides and rear - and the painted body leaves through the exit. Every motion is planned with collision checking and executed by the robot controllers in a physics simulation. The final run reaches 98.4 % paint coverage in a 16.5-minute cycle.



(a) cell in Gazebo *(b) top view*

Figure 1 - The paint cell: booth with entry and exit zones, skid conveyor, loop track and robot.

2. Software

Tool	Use in the project
ROS 2 Jazzy	framework: robot description, nodes, launch, communication
MoveIt 2	inverse kinematics, motion planning (Pilz PTP, OMPL), collision checking
Gazebo Harmonic + ros2_control	physics simulation, joint trajectory controllers, visualisation
RViz 2	planning, collision checks and paint markers
SolidWorks	CAD of the skid, track, carriage, spray gun and car preparation
Python (numpy, scipy)	paint path planner, coverage simulation, result figures

An open, industry-standard robotics stack was chosen so the robot is driven by real planners and controllers, not by a scripted animation; the same planner runs on mock hardware or in Gazebo.

3. Robot and tool

The robot is a KUKA KR 210 R3100-2 (6 axes, 3.1 m reach), using KUKA's official ROS 2 model. It stands on a carriage on a 2-axis floor track that acts as a 7th axis, so one robot can reach every side of the car. The spray gun is mounted on the flange; the tool point (TCP) is defined 0.25 m in front of the nozzle, so keeping the TCP on the surface keeps the spray stand-off constant. In production a dedicated, explosion-proof paint robot would be used; the KR 210 was chosen for its reach and its accurate official model.

4. Cell layout

The layout follows the reference drawing (booth between an entry and an exit zone, three rooms along one side), but the booth was enlarged from 8 x 5 m to 9.5 x 6.5 m. A reach study showed that with the track 2.27 m from the car centre line the lower door area was out of reach; at 2.5 m every target was reached with a minimum clearance of 226 mm. The track loop (8 x 5 m centre line) and the conveyor then require the larger booth.

- **Conveyor:** roller line through both doors; the car sits on a skid whose four posts carry the body sills. Transfer gaps let the robot carriage cross the car path at both ends of the track.

- **Track:** two prismatic axes (x and y). Joint 1 of the robot is limited to +-185 deg and cannot turn towards the walls (0.75 m behind the track), so the robot does not circle the car: it works front -> near side -> rear -> far side and returns the same way.
- **Booth walls** are part of the MoveIt planning scene, so no motion can be planned through them.

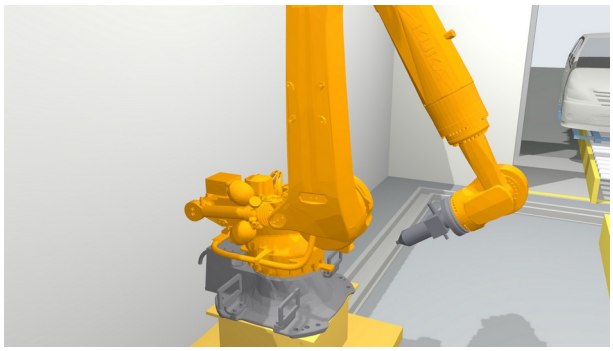


Figure 2 - KR 210 on its track carriage with the spray gun.

5. Approach

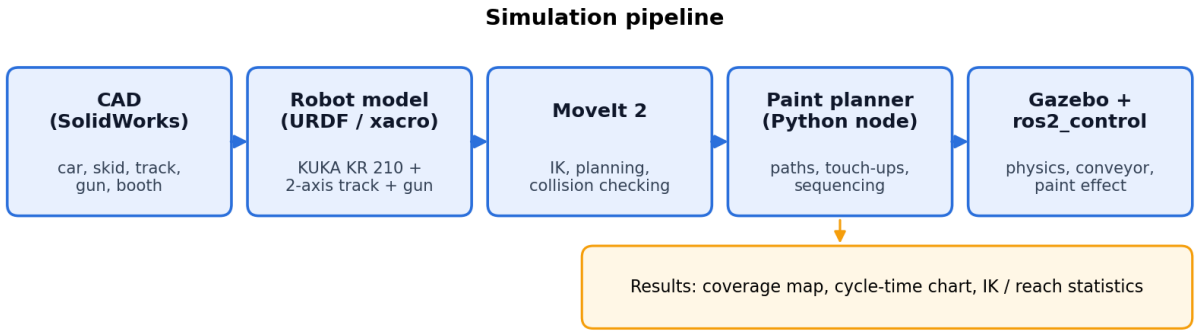


Figure 3 - Simulation pipeline.

The car model was cleaned in CAD, scaled to 4.80 x 1.85 x 1.60 m and split into six paint panels; window glass and trim are not painted. A paint job is defined per panel and track side (8 jobs in total). All trajectories are planned first; then the car enters, the robot executes the jobs, and the car leaves. The paint shown in the simulation is exactly the area hit by the spray, which also gives the coverage.

Challenges solved:

- **Unreachable centre line:** the roof and hood are split at y = 0 and painted half from each side.
- **Arm folded against a joint limit near the doors:** the carriage runs 1.2-1.5 m behind the gun, so the robot paints diagonally (reachability on the sides rose from 85-96 % to over 99 %).
- **Joints sticking at their end stops in simulation:** soft joint limits and wrist angles kept mid-range.
- **Low simulation speed:** paint patches merged and the physics step set to 4 ms (real-time factor 0.4 -> 1.0).

6. Trajectory and path planning

- **Paint path:** parallel rays are cast onto each panel along the spray direction. This gives zig-zag stripes 0.15 m apart (50 % overlap of a 0.30 m spray fan) with a waypoint every 0.05 m. The gun follows the smoothed surface normal (tilt at most 20 deg); windows are crossed with the spray off.
- **7-axis motion:** the track follows the gun along each stripe while MoveIt's collision-aware IK solves the arm for every waypoint, seeded with the previous solution for continuous motion.
- **Timing:** a constant gun speed of 0.4 m/s, slowed down wherever a joint velocity or acceleration limit would be exceeded (smooth stops and stripe turn-arounds).
- **Approach / retract:** every pass starts and ends 0.4 m back along the gun axis, clear of the car.
- **Touch-ups:** the spray of the planned passes is simulated; the spots it misses (curved edges, corners) get short extra strokes with up to 50 deg gun tilt.
- **Moves between passes:** straight joint moves (Pilz PTP); OMPL finds a detour when the straight move is blocked.

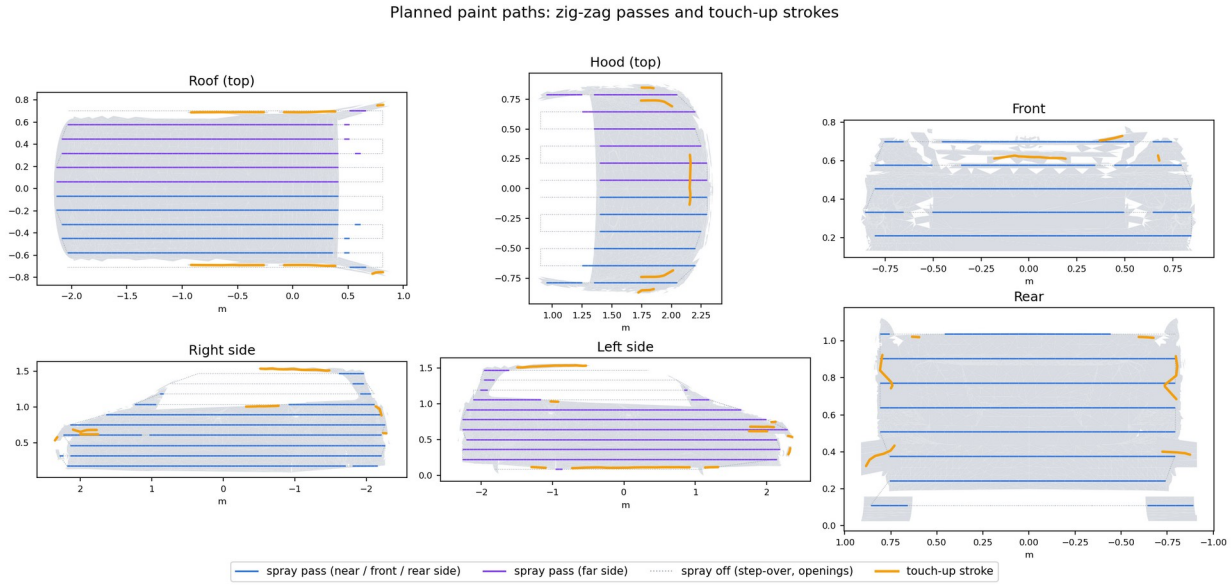
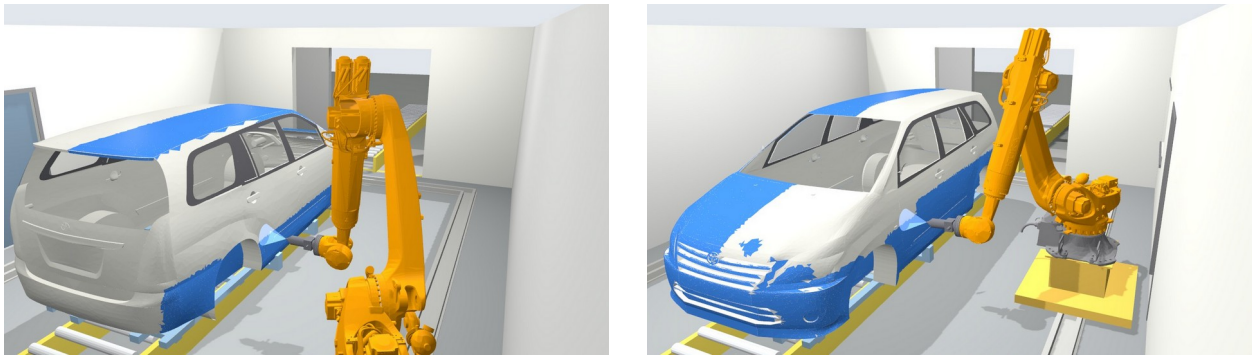


Figure 4 - Planned paths: zig-zag passes (blue: near / front / rear side, purple: far side), spray-off moves (dotted) and touch-up strokes (orange).

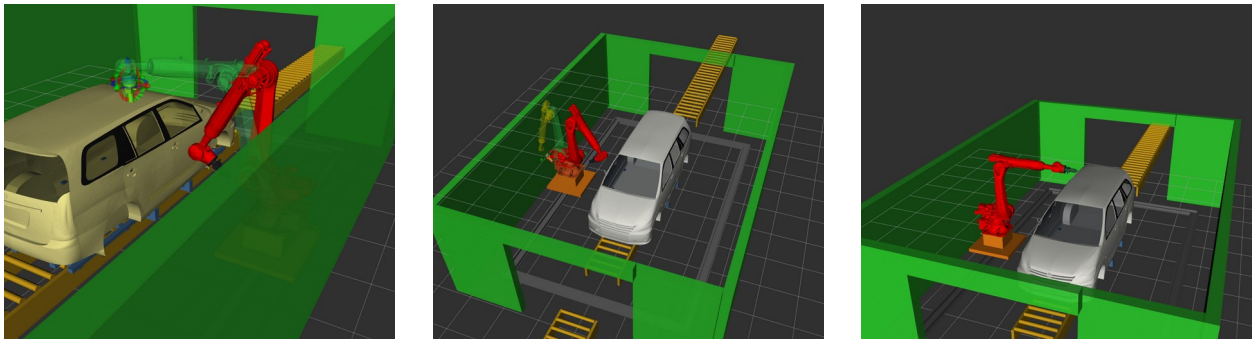


(a) right side, spray plume visible *(b) left side from the far track*

Figure 5 - Painting in Gazebo: only the sprayed area turns blue.

7. Collision checking

MoveIt checks the robot and spray gun against the car body, skid, conveyor, track and booth walls, and the robot against itself (the self-collision matrix was built from 3,000 random robot poses). The check runs for every IK waypoint of every paint pass and along every planned move; a pose or path in collision is rejected and the real robot does not move. Soft joint limits keep all motion 0.06-0.3 rad away from the mechanical stops.



(a) gun into the roof: rejected *(b) arm into the wall: rejected* *(c) free pose: executed*

Figure 6 - Collision checking in RViz: the transparent robot is the requested goal; colliding parts are highlighted and planning fails.

8. Results

Panel	Area (m2)	Stripes	IK solved	Touch-ups reached	Passes only (plan)	Coverage
Front	1.01	5	100.0 %	3 / 3	97.8 %	99.9 %
Hood	2.04	12	100.0 %	3 / 5	96.6 %	98.8 %
Roof	3.62	12	100.0 %	6 / 6	98.8 %	100.0 %
Right side	5.02	10	99.5 %	4 / 7	96.0 %	98.1 %
Rear	1.93	8	98.7 %	4 / 6	93.9 %	95.9 %
Left side	4.96	11	99.7 %	7 / 10	94.7 %	97.9 %
Total	18.6	58				98.4 %

The robot sprayed 127 m of path. The cycle took 990 s of simulated time: painting 498 s, touch-ups 109 s, approach / retract 143 s, travel 151 s and conveyor 86 s. The touch-ups raised the coverage from about 96 % (passes only) to 98.4 %. What remains unpainted is mainly the rear bumper corners, which wrap round to the sides and cannot be reached from the rear track; 30 of 37 touch-up strokes were reachable.

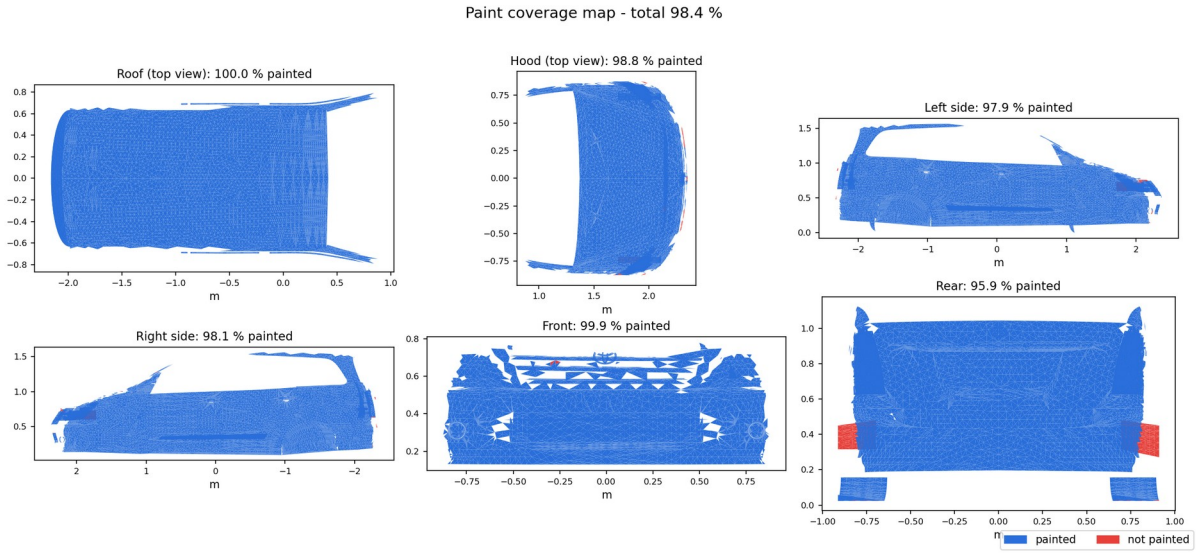


Figure 7 - Coverage map of the final run (blue painted, red not painted).

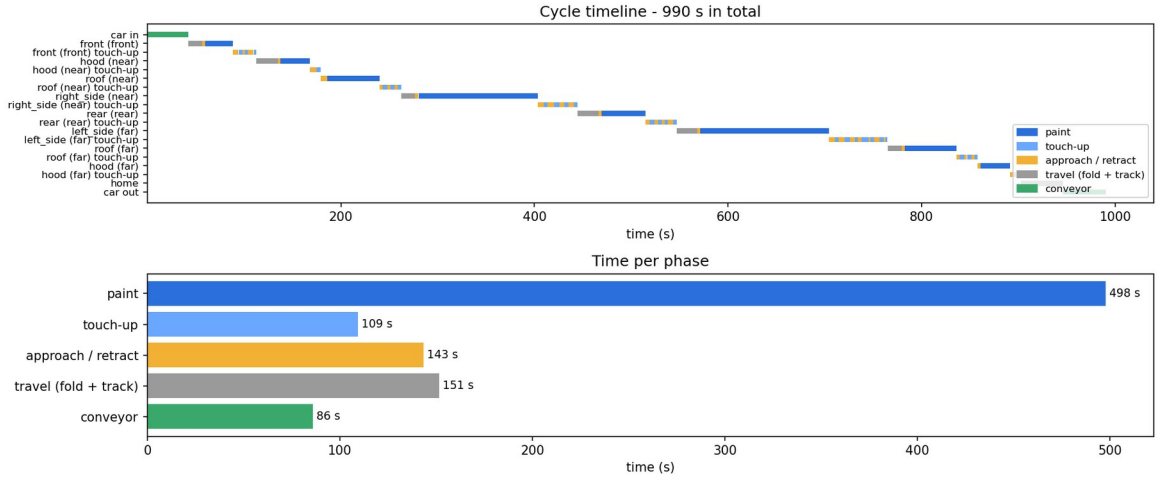


Figure 8 - Cycle timeline and time per phase.

9. Limitations and future work

- Paint is modelled as coverage, not film thickness; a deposition model (spray profile, overlap, speed) would allow thickness maps and speed optimisation.
- A dedicated paint robot, and two robots (one per side), would shorten the cycle substantially.
- The track rails would be recessed where they cross the conveyor; the rear bumper corners need an extra robot position or a different track layout.